ENTERPRISE PRODUCT SPECIFICATION

ATTENDORA — PRODUCT REQUIREMENTS DOCUMENT

Multi-Tenant Cloud-Based Student Attendance Management & Compliance System

Document ID: ATD-PRD-V1.0
Status: Approved for Baseline
Security: Institutional Confidential

1. DOCUMENT CONTROL & STAKEHOLDERS

This Product Requirements Document (PRD) defines the absolute product baseline, technical surface, functional mechanics, acceptance bounds, and release boundaries for Attendora. It serves as the single source of truth for Product, Engineering, Quality Assurance, UI/UX, and Executive Stakeholders.

1.1 Document Metadata

Attribute	Specification Details
Product Name	Attendora (Cloud-Based Student Attendance Management System)
Document Version	1.0.0 (Production Master)
Document Owner	Principal Product Lead (EdTech Platforms Group)
Target Baseline	Q3–Q4 Academic Institutional Onboarding Cycle
Target Audience	Core Engineering (Backend, Web Frontend), QA Automation, UX Design, Academic Operations, Department Heads

1.2 Revision History

Version	Date	Author	Summary of Changes
v0.1.0	12 Oct 2026	Lead PM / Founders	Initial product draft outlining core problem statement and technology feasibility.
v0.5.0	24 Oct 2026	Staff Product Manager	Added comprehensive RBAC matrices, database entity schemas, and edge case specifications.
v1.0.0	04 Nov 2026	Enterprise PRD Review	Approved baseline specifications, fully defined acceptance criteria, edge condition rules, and NFRs.

1.3 Stakeholders & Governance Sign-Off

Role	Responsibility	Sign-Off Authority	Status
Lead Product Manager	Feature definition, roadmaps, scope boundaries, acceptance approval	Product Operations Lead	**Approved**
Technical Architect	Flask application design, Firebase rules, deployment infra & scalability	Engineering VP / Tech Lead	**Approved**

Role	Responsibility	Sign-Off Authority	Status
Lead UI/UX Designer	User interaction flows, responsive layout specs, error states	Design Principal	**Approved**
Quality Assurance Lead	End-to-end test harnesses, edge condition verifications, regressions	Director of Quality Engineering	**Approved**
Institutional Operations	Academic compliance, timetable compatibility, faculty onboarding	Academic Dean Advisory Group	**Approved**

> **PRIORITY CLASSIFICATION FRAMEWORK**
> All requirements inside Section 9 are strictly cataloged using standard priority tokens:
> **P0 (Critical / Blocker):** Core baseline mandatory for MVP deployment. The system cannot go live without 100% test coverage and zero outstanding severity-1 bugs.
> **P1 (High Priority / Fast Follow):** Vital operational enhancements scheduled immediately following initial user pilot stabilization.
> **P2 (Desirable / Future Phase):** Advanced automations, third-party SIS integrations, and biometrics reserved for v2.0+ roadmap milestones.

2. INTRODUCTION & PRODUCT VISION

2.1 Purpose & Execution Intent

Paper registers, fragmented desktop spreadsheets, and legacy offline software cripple academic institutions with manual tally errors, proxy attendance, and unacceptably delayed low-attendance interventions. Attendora provides an authoritative, browser-based, multi-tenant digital attendance infrastructure that enables faculty to record lecture records in seconds, grants students immediate visibility into their attendance thresholds, and equips institutional administrators with real-time statutory compliance reporting.

2.2 Product Vision Statement

"To establish the most frictionless, transparent, and dependable attendance tracking ecosystem for higher education institutions—eradicating administrative friction, safeguarding academic compliance, and providing instantaneous actionable insights to students and faculty."

2.3 Executive Product Summary

Attendora is an enterprise-ready, cloud-native Software as a Service (SaaS) web application built with a responsive Bootstrap frontend, Python Flask RESTful backend, and real-time cloud data store, engineered to replace manual roll-call ledgers with secure, instant, and auditable digital attendance lifecycles.

2.4 Product Boundary & Scope Definition

This document governs the end-to-end functional criteria, data definitions, security constraints, and operational boundaries of the Attendora web system. The boundaries are specified as follows:

- **In-Scope Boundaries:** Web application workflows for Students, Faculty/Teachers, and Administrators; institutional multi-department hierarchy; manual digital roll-call and bulk-mark mechanics; student self-service percentage tracking; statutory low-attendance alerts (75% regulatory limit); Excel/CSV and PDF institutional report generation; role-based access control and secure authentication.
- **Out-of-Scope (Excluded from Current Baseline):** Direct biometric/RFID hardware sensor firmware integration; automatic native payment gateway processing for student fines; complete learning management systems (LMS) assignments/grades; physical room door-lock automation.

3. PROBLEM STATEMENT & MARKET CONTEXT

3.1 Problem Statement by User Persona

- **For Faculty & Lecturers:** Taking roll-call on paper ledgers consumes 10–15 minutes of every 60-minute instructional session. Tallying physical attendance sheets at the end of the term requires 40+ hours of tedious calculation, producing severe transcription and calculation errors.
- **For Students:** Students lack continuous visibility into their cumulative and subject-level attendance. They only discover they have breached mandatory minimum compliance thresholds (typically 75% under university standards) when debarred right before final semester examinations, preventing corrective intervention.
- **For Academic Administrators & Deans:** Department heads have zero real-time visibility into lecture conduction status or daily absenteeism patterns. Compiling institutional reports for regulatory accreditations requires weeks of manual data aggregation across dozens of physical binders.

3.2 Current Alternatives & Institutional Baselines

Alternative	Mechanism & Operational Reality	Systemic Deficiencies
Physical Roll-Call Registers	Paper ledgers manually checked by faculty per class period and submitted to departmental offices monthly.	Vulnerable to physical loss, page destruction, proxy marking, unsearchable records, and no student visibility.
Spreadsheets (Excel / Sheets)	Faculty maintain individual offline Excel workbooks uploaded to shared network drives or emails.	Formula corruption, severe version divergence, unauthenticated overwrites, and fragmented student records.
Monolithic ERP Systems	Legacy, on-premise client-server suites installed on local college intranet computers.	Exorbitant licensing costs, rigid configurations, non-responsive mobile views, and frequent downtime.

3.3 Pain Points & Economic Costs

- **Direct Productivity Waste:** An institution with 80 faculty members conducting 4 lectures daily loses approximately 53 faculty hours per day strictly performing physical roll-calls.
- **Statutory & Legal Exposure:** Inaccurate attendance calculation leads to wrongful student debarment disputes, parental appeals, and accreditation compliance sanctions from higher education bodies.
- **Data Latency:** Traditional attendance consolidation runs on a 30-day billing or monthly cycle, rendering timely behavioral or academic intervention impossible.

3.4 Why Now

Educational institutions in India and global emerging markets are executing rapid digital transformations. With ubiquitous 4G/5G mobile connectivity across campuses and faculty using personal smartphones and laptops in classrooms, a cloud-first, mobile-responsive web application eliminates the capital expense barrier of expensive dedicated scanning hardware.

3.5 Competitive Landscape & Differentiation Matrix

Dimension	Legacy ERPs	Biometric / RFID Vendors	Attendora SaaS
Deployment Speed	3–6 Months (Complex server setups)	4–8 Weeks (Hardware wiring/cabling)	**Instant Self-Serve / < 24 Hours**
Hardware Investment	High local server costs	Expensive sensor units per doorway	**Zero Hardware (BYOD Web)**
Ease of Use	Cluttered, 90s-style legacy UIs	Firmware dependencies, card losses	**Clean, 3-click Responsive UI**
Student Access	Rarely exposed outside campus LAN	Terminal screen or physical kiosk	**24/7 Cloud Access on any device**

4. GOALS, OBJECTIVES & NON-GOALS

4.1 Business Goals

Goal ID	Strategic Business Goal	Measurable Target	Timeframe
BG-01	Establish initial institutional customer foothold	15 Higher Education Colleges onboarded	First 6 Months post-launch
BG-02	Achieve sustained operational engagement	> 90% Faculty Monthly Active Usage	Month 3 onwards per college
BG-03	SaaS Subscription Conversion (Freemium to Paid)	25% Tier-1 institutional pilot conversion	Month 9 post-pilot
BG-04	Low System Churn Rate	< 5% annual institutional contract churn	Annual renewal cycle

4.2 Product Goals

- **Radical Roll-Call Acceleration:** Reduce the time required for a faculty member to mark attendance for a 60-student class from 12 minutes to under 45 seconds.
- **Complete Information Transparency:** Ensure 100% of enrolled students have real-time visibility into their overall and subject-level attendance percentages within 2 seconds of faculty submission.
- **Instant Institutional Auditing:** Allow Academic Administrators to generate verified, printable compliance reports across any department, semester, or section in less than 5 clicks.

4.3 Explicit Non-Goals (Scope Fence)

- **No Automated Biometric Hardware Control:** The system will not directly communicate with physical biometric USB devices, fingerprint scanners, or RFID card readers in Version 1.0.
- **No Examination Grading or LMS Workflow:** Attendora will not manage assignments, grades, exam transcripts, or course video streaming; it is strictly an attendance and compliance engine.
- **No Financial Fee Collection:** Attendora will calculate attendance-shortage fines where applicable, but will not act as a credit card payment gateway or merchant account processor for student fees in MVP.

5. SUCCESS METRICS & KPIS

5.1 North Star Metric

> **NORTH STAR METRIC: VERIFIED RECORDED STUDENT-LECTURES (VRSL)**
> **Definition:** The total number of unique student-lecture sessions successfully recorded, validated, and finalized into the database per academic day.
> **Strategic Justification:** This metric captures core customer value across all three personas: faculty actively using the system, students receiving tracking entries, and institutions establishing regulatory audit trails.

5.2 Primary KPIs (AARRR Framework Aligned)

Lifecycle Stage	Key Performance Indicator	Operational Definition	Target Objective
Acquisition	Institution Roster Completion	% of invited colleges uploading full student and faculty rosters within 7 days.	> 80% onboarding completion
Activation	First-Lecture Execution Rate	% of onboarded faculty who successfully mark their first digital lecture within 48h.	> 85% activation
Retention	Weekly Active Faculty (WAF)	Faculty logging in and submitting ≥ 4 lecture attendance rosters per week.	> 92% sustained retention
Engagement	Student Portal Session Rate	Average monthly logins per enrolled student reviewing status.	> 4.0 sessions/month/student
Quality / SLA	Submission Error Rate	Percentage of attendance submission API requests resulting in HTTP 5xx errors.	< 0.05% of all submissions

5.3 Guardrail Metrics

The engineering and product teams must track these metrics to ensure rapid product iteration does not degrade product integrity:

- **Data Revision Dispute Rate:** Must remain below 0.2% of total attendance marks logged (ensures teacher accuracy and low tampering).
- **Critical Path API Latency:** 95th percentile latency for roll-call submission must stay < 500ms on standard 3G/4G connections.
- **System Availability:** Core service uptime must maintain a strict 99.9% availability during institutional operating hours (07:00 to 20:00 IST).

6. TARGET USERS & PERSONAS

Persona 1: Prof. Rajesh Deshmukh — Assistant Professor (Faculty Member) PRIMARY OPERATIONAL USER

Context & Environment: 42 years old, teaching Computer Science at an engineering college in Pune. Conducts 3 lectures and 2 practical lab sessions daily across different classrooms. Uses a mid-range Android smartphone and an institutional Windows laptop.

Core Objectives: Complete lecture attendance within 60 seconds without disrupting classroom momentum; maintain error-free records; quickly identify chronically absent students before midterm exams.

Frustrations: Spends up to 15 minutes manually calling names from paper registers; dealing with proxy roll-calls; spending entire weekends manually tallying monthly percentages for departmental office submissions.

Device Context: Mobile Chrome browser on Android inside classrooms over campus Wi-Fi / 4G cellular; desktop browser in faculty cabin.

Persona 2: Sneha Kulkarni — 3rd Year B.Tech Student (Student) CONSUMER USER

Context & Environment: 20 years old, balancing heavy academic coursework, project assignments, and campus placements. Uses an iPhone / Android smartphone constantly on the move.

Core Objectives: Always know exact real-time attendance percentage across all theory subjects; receive immediate alerts when approaching the dangerous 75% boundary; verify absence records to detect errors.

Frustrations: Finding out only a week before semester exams that an unexpected absence resulted in debarment; having no proof when a teacher marked them absent by mistake.

Device Context: 95% mobile web browser access via personal smartphone.

Persona 3: Dr. Anita Roy — Academic Dean & Department Administrator (Admin) ADMINISTRATIVE & GOVERNANCE USER

Context & Environment: 54 years old, responsible for academic compliance, regulatory university audits, faculty oversight, and parental escalations for an institute of 2,400 students.

Core Objectives: Instantly view institution-wide attendance health; export verified compliance rosters for University / Accreditation inspections; enforce accountability among teaching staff.

Frustrations: Missing, illegible, or modified paper logs; delays in receiving departmental registers; handling angry parents whose children were debarred without documented advance notice.

Device Context: Desktop workstation (Windows/macOS), high-resolution widescreen display.

7. USER STORIES & JOBS TO BE DONE (JTBD)

7.1 Jobs to Be Done (JTBD) Framework

- **Faculty JTBD:** *"When I step into the lecture hall, I want to quickly capture student presence with minimal clicks, so that I can maximize actual teaching time and avoid manual administrative calculation work."*
- **Student JTBD:** *"When I am planning my academic schedule, I want to see my real-time verified attendance percentage per subject, so that I can remain compliant with institutional requirements and avoid sudden debarment."*
- **Administrator JTBD:** *"When regulatory inspections or audit cycles occur, I want to generate authoritative, auditable attendance reports across all departments, so that I can guarantee statutory compliance without staff overhead."*

7.2 User Stories by User Type

User Role	Capability Desire	User Story Statement	Core Value
Faculty	Default Present Bulk Mark	*As a teacher, I want all students to be marked 'Present' by default so that I only need to tap the names of the few absent students.*	Speed & Efficiency
Faculty	Session Editing Window	*As a teacher, I want to correct an accidental attendance entry within a restricted time window so that my records stay strictly accurate.*	Error Correction
Student	Deficit Margin Calculator	*As a student, I want to see how many future consecutive lectures I must attend to recover above 75% so that I can take corrective action.*	Actionable Guidance
Student	Detailed Date Log	*As a student, I want to inspect the exact calendar dates and time slots when I was marked absent so that I can contest errors immediately.*	Transparency
Admin	Bulk Roster Ingestion	*As an admin, I want to upload CSV files of students and faculty so that I can set up an entire department in minutes without manual data entry.*	Rapid Onboarding
Admin	Low-Attendance Defaulters List	*As an admin, I want to filter and export all students with < 75% attendance across any semester so that formal warning notices can be issued.*	Compliance & Control

8. SCOPE, ASSUMPTIONS, CONSTRAINTS & DEPENDENCIES

8.1 Scope Boundaries

- **Mandatory In-Scope:** Multi-role authentication (Student, Faculty, Admin); responsive mobile-first digital roll-call sheet; real-time attendance calculation engine; threshold breach flagging (75% default); printable and downloadable PDF/Excel consolidated summaries; department, class, and subject configuration consoles.
- **Explicitly Out of Scope:** Automated GPS-geofenced student self-check-in; facial recognition video feed inference; fee management payment gateways; multi-school district hierarchical payroll processing.

8.2 Documented Assumptions

Assumption ID	Statement of Operational Assumption	Risk / Impact if Invalid
ASM-01	Institutions possess reliable Wi-Fi or cellular data (3G/4G/5G) across lecture halls.	Offline roll-call local caching would be required immediately rather than in Phase 2.
ASM-02	Colleges enforce a standard 75% minimum attendance rule based on state university mandates.	System must support configurable institutional thresholds at the admin settings tier.
ASM-03	Faculty members possess personal smartphones or laptops in classrooms (BYOD model).	Low adoption if college must supply dedicated hardware terminals to staff.

8.3 Project Constraints

- **Technical Constraints:** Application backend built in Python Flask; client interface styled with lightweight Bootstrap without heavy node-based SPA frameworks to maximize compatibility with low-end mobile devices; data storage managed on cloud infrastructure (Firebase / PostgreSQL on Render/AWS).
- **Resource & Budget Constraints:** Initial architecture calibrated for minimal hosting overhead, requiring database queries and bandwidth to fit within high-efficiency parameters.
- **Regulatory & Privacy Constraints:** Compliance with Indian Digital Personal Data Protection (DPDP) Act; student contact numbers and records must be protected behind strict institutional tenant barriers.

8.4 System Dependencies

Dependency Entity	Type	Description	Impact if Delayed
Cloud Database (Firebase/Firestore)	Infrastructure	Persistent cloud document/relational store for real-time reads and writes.	System blocked; no state persistence.
Deployment Host (Render / AWS)	PaaS / Cloud	Containerized Python WSGI runtime for web application delivery.	Inability to serve production endpoints.
Institutional CSV Datasets	Customer Asset	Standardized student roll number, email, and class rosters provided by colleges.	Delayed institutional onboarding.

9. FUNCTIONAL REQUIREMENTS

The following tables specify the mandatory capabilities of the Attendora system. Each requirement is assigned a deterministic ID, strict priority, and independently verifiable acceptance criteria.

9.1 Module: Authentication, Identity & Session Management (AUTH)

Req ID	Feature Description	Priority	Testable Acceptance Criteria
FR-AUTH-01	Multi-Role Secure Login	P0	1. System provides email/username and password entry fields. 2. Upon valid credential submission, user is authenticated and redirected to role-specific dashboard (Student, Faculty, Admin) within 1.5s. 3. Invalid credentials trigger inline error: *"Invalid username or password"* without revealing account existence. 4. Rate limits to max 5 failed attempts per IP per 10 minutes before locking out for 15 minutes.
FR-AUTH-02	Role-Based Access Control (RBAC) Enforcement	P0	1. Unauthenticated users accessing any internal route are redirected to `/login` with 302 code. 2. Students attempting to access `/faculty/*` or `/admin/*` routes receive HTTP 403 Forbidden with zero data leaked. 3. Faculty attempting to access `/admin/*` receive HTTP 403 Forbidden. 4. User session state is validated on every server transaction.
FR-AUTH-03	Session Expiry & Secure Logout	P0	1. User sessions expire after 8 hours of inactivity. 2. Clicking "Log Out" invalidates the server-side session token immediately. 3. Browser back-navigation post-logout displays login screen and does not display cached private data.
FR-AUTH-04	Password Reset Workflow	P1	1. User enters registered email; system sends cryptographically random, single-use reset token valid for 30 minutes. 2. Token expires immediately upon successful password update. 3. Passwords must enforce minimum 8 characters with at least 1 number and 1 special symbol.

9.2 Module: Faculty Operations & Attendance Marking (FAC)

Req ID	Feature Description	Priority	Testable Acceptance Criteria
FR-FAC-01	Lecture Session Creation	P0	1. Faculty selects Department, Class/Semester, Section, and assigned Subject from dropdowns. 2. Date defaults to current calendar day; lecture time-slot defaults to current hour. 3. System validates that the faculty member is officially assigned to teach the selected subject. 4. Duplicate lecture prevention: If a lecture record exists for the exact same subject, section, date, and hour, system prompts confirmation: *"A lecture already exists for this slot. Overwrite or create extra session?"*
FR-FAC-02	Interactive Digital Roll-Call Sheet	P0	1. System dynamically renders table containing all active enrolled students sorted by Roll Number. 2. Every student displays: Roll Number, Full Name, and a toggleable status badge. 3. **"Mark All Present" default:** All toggles initialize in "Present" state. 4. Single tap/click on student row flips state to "Absent" (highlighted in high-contrast red). 5. Persistent sticky footer displays dynamic count: *"Total: X | Present: Y | Absent: Z"* updated instantaneously on client side.
FR-FAC-03	Attendance Submission & Finalization	P0	1. Clicking "Submit Attendance" triggers modal summary showing counts. 2. Confirmation commits atomic batch write to cloud database. 3. System generates unique `SessionID`, timestamp, and audit trail of user ID. 4. Successful write redirects to confirmation view within 1.0s and shows success alert banner.
FR-FAC-04	Attendance Modification Grace Period	P0	1. Faculty can edit a submitted attendance sheet within a strict **24-hour window** post-submission. 2. Any alteration after submission requires an explicit textual "Reason for Modification" entry (minimum 10 characters). 3. Modifying entries updates the student records and appends an immutable record to the Audit Log. 4. Post 24 hours, the record locks permanently for faculty and requires Admin elevation to modify.
FR-FAC-05	Class-Level Attendance Summary View	P1	1. Faculty can view monthly cumulative percentage grid for all students in their assigned subjects. 2. Rows with cumulative percentage < 75% are visually highlighted in light red. 3. Clicking any student expands the chronological history of all conducted lectures and presence status.

9.3 Module: Student Self-Service Dashboard (STU)

Req ID	Feature Description	Priority	Testable Acceptance Criteria
FR-STU-01	Cumulative Attendance Overview	P0	1. Student dashboard prominently displays overall institutional attendance percentage calculated via formula: `(Total Classes Attended / Total Classes Conducted) * 100`. 2. Displays total attended and total conducted counts. 3. Color coded: Green (≥ 75%), Amber (70.0%–74.9%), Red (< 70.0%). 4. Data updates immediately when a faculty member submits a new session.
FR-STU-02	Subject-Wise Breakdown Cards	P0	1. Lists all enrolled courses for the active semester. 2. Each card shows: Course Name, Course Code, Faculty Name, Classes Attended / Conducted, and Percentage rounded to 1 decimal place. 3. Includes status badge: "Safe", "At Risk", or "Critical Shortage".
FR-STU-03	Shortage Recovery Calculator	P1	1. For any subject where percentage is < 75%, system displays dynamic calculation: *"You must attend next N consecutive classes to reach 75%."* 2. Mathematical model: `ceil((0.75 * Conducted - Attended) / 0.25)`. 3. If percentage is ≥ 75%, displays: *"You can afford to miss M upcoming classes while maintaining 75%."*
FR-STU-04	Date-Wise Historical Attendance Ledger	P1	1. Interactive log listing all class sessions held for the student's section. 2. Filters by Date Range and by Subject. 3. Explicitly shows: Date, Time Slot, Subject, Faculty Name, Status (Present/Absent). 4. Includes "Dispute Status" button to generate formal error inquiry email to the faculty.

9.4 Module: Administrative Management & Institutional Reporting (ADM)

Req ID	Feature Description	Priority	Testable Acceptance Criteria
FR-ADM-01	Batch Student & Faculty Onboarding (CSV)	P0	1. Admin can upload `.csv` file containing: `[RollNo, FullName, Email, Department, Year, Section]`. 2. System performs pre-validation: flags missing headers, malformed email syntaxes, or duplicate roll numbers before writing. 3. Displays preview table with valid rows and flagged errors. 4. Commits valid records and generates default account credentials for batch users.
FR-ADM-02	Academic Structure Configuration	P0	1. Admin creates, edits, and disables: Departments, Academic Years, Semesters, Sections, and Subjects. 2. Maps Faculty to specific Subjects and Sections. 3. Prevents deletion of any subject or section that has active attendance records attached to preserve relational integrity.

Req ID	Feature Description	Priority	Testable Acceptance Criteria
FR-ADM-03	Defaulters / Low-Attendance Report Generator	P0	1. Admin selects Department, Semester, and customizable percentage threshold (default: 75%). 2. System queries all enrolled students and aggregates records across all subjects. 3. Outputs list of students falling below threshold, showing individual subject percentages and overall aggregate. 4. Supports one-click export to formatted Excel (`.xlsx`) and PDF document suitable for printing.
FR-ADM-04	Attendance Record Override & Audit	P1	1. Admin can override historical attendance records beyond the faculty 24-hour lock window (e.g., approved medical leave). 2. Form requires: Approver Name, Medical/Event Certificate Reference ID, and Justification note. 3. Original status, modified status, timestamp, and Admin ID logged irreversibly in Audit Log.
FR-ADM-05	Institutional Analytical Dashboard	P1	1. Dashboard shows institution-wide metric tiles: Total Lectures Scheduled Today, Total Lectures Marked, Overall Daily Attendance Rate (%). 2. Department-wise comparison bar table showing relative compliance. 3. List of missing roll-calls: highlights faculty members who had scheduled classes but have not submitted attendance.

9.5 Module: Alerts & Notification Engine (NOTIF)

Req ID	Feature Description	Priority	Testable Acceptance Criteria
FR-NOTIF-01	Low-Attendance Dashboard Alerts	P0	1. When a student's cumulative subject percentage drops below 75.0%, an persistent alert banner appears on their dashboard. 2. Banner states: *"Warning: Your attendance in [Subject Name] has fallen to [X]%. Minimum requirement is 75%."* 3. Banner cannot be dismissed until attendance recovers above threshold.
FR-NOTIF-02	Missing Lecture Reminder for Faculty	P1	1. If a timetable session ends and attendance has not been submitted within 60 minutes, system displays a prompt on faculty login. 2. Email reminder dispatched at 17:00 daily listing any unsubmitted classes for that day.

10. EDGE CASES & EXCEPTION HANDLING

To guarantee system resilience in realistic educational environments (flaky campus Wi-Fi, accidental browser closes, late admissions), the system must handle exceptions predictably.

Failure Scenario	Underlying Condition / Trigger	Prescribed System Behavior & Recovery
Mid-Roll-Call Network Drop	Teacher marks 40 students; device loses internet connectivity before hitting "Submit".	Client-side local storage retains toggled states. UI alerts: *"Connection lost. Changes cached locally. Do not close tab."* Auto-retries submission when connection restores.
Concurrent Double-Submission	Teacher accidentally double-taps the "Confirm Submit" button rapidly on mobile.	Client disables submit button immediately on first tap (idempotency key generated). Backend validates unique combination of `(SubjectID + SectionID + Date + Slot)`; second request is rejected gracefully.
Student Added Mid-Semester	A student joins class in week 6 after 20 lectures have already been conducted.	Admin enrolls student with an "Enrollment Effective Date". Attendance calculation formula ignores lectures conducted prior to effective date; student is not penalized for prior sessions.
Zero Lectures Conducted (Div/0)	New semester starts; student or subject dashboard loaded before any attendance is logged.	System detects `Total Conducted == 0`. Percentage displays `"N/A"` or `"100.0% (No classes held)"`; prevents division-by-zero application crashes.
Unauthorized Direct URL Probe	Student attempts typing `/admin/export-defaulters` directly into browser bar.	Backend auth middleware intercepts request, confirms absence of Admin claims, logs security warning event, and issues HTTP 403 redirect to `/student/dashboard`.
Corrupt CSV Ingestion	Admin uploads a CSV file containing 500 rows, but row 42 has a missing email and row 108 has an invalid character.	Atomic transaction dry-run: System processes zero writes, displays validation error modal showing exact row numbers and issues (*"Row 42: Email missing"*), and prompts re-upload.

11. KEY USER FLOWS

11.1 Primary Faculty Roll-Call Flow (Happy Path & Correction)

> **Step 1: Selection & Initialization**
> Faculty navigates to `/faculty/mark` → Selects Class (e.g., TE-Computer), Section (A), and Subject (Database Management Systems) → Date auto-populates to Today → Clicks *"Load Roster"*.

> **Step 2: Interactive Marking**
> System renders 60 enrolled students. All initialize to **"Present"** → Faculty calls roll or observes classroom → Taps rows for 4 absent students → Badges change to red "Absent" → Counters display *56 Present / 4 Absent*.

> **Step 3: Verification & Submission**
> Faculty taps *"Submit Attendance"* → Confirmation popup summarizes figures → Faculty clicks *"Confirm & Save"* → API executes POST request → System commits record → Success banner displays with option to download immediate PDF summary.

Step 4: Post-Submission Correction (Within 24h)
Student arrives late with official permission → Faculty opens *"Recent Sessions"* → Selects session → Changes student from "Absent" to "Present" → Enters reason: *"Late arrival with Dean permission"* → Clicks *"Update Record"* → Audit trail appends modification.

11.2 Student Attendance Verification Flow

Step 1: Dashboard Login
Student enters credentials at `/login` → Redirects to `/student/dashboard`.

Step 2: Threshold Inspection
Student reviews overall aggregate banner (e.g., 72.4% - Amber Warning) → Inspects subject card for "Data Structures" showing 68.0% (Critical Shortage).

Step 3: Calculator Analysis & Date Drill-down
Student views recovery recommendation: *"Attend next 5 consecutive lectures to reach 75.0%"* → Clicks *"View Session History"* to verify dates marked absent → Confirms legitimacy of records.

11.3 Admin Institutional Reporting Flow

Step 1: Report Generation
Admin enters `/admin/reports` → Selects Department: "Mechanical Engineering", Semester: "Sem 6", Threshold: "75%" → Clicks *"Generate Compliance Report"*.

Step 2: Inspection & Export
System displays table of 22 flagged defaulters → Admin verifies details → Clicks *"Export Formal PDF Report"* → Browser downloads print-ready institutional document formatted with official university header.

12. NON-FUNCTIONAL REQUIREMENTS (NFRS)

The system must comply with rigorous operational benchmarks to ensure high reliability in academic campus settings.

NFR ID	Category	Requirement Specification	Target Acceptance Metric
NFR-PERF-01	Performance	Class roster loading time for 100 students on mobile devices.	< 800ms over 4G cellular connections.
NFR-PERF-02	Performance	Batch attendance write execution latency (commit to database).	< 500ms p95 latency.
NFR-SCAL-01	Scalability	Concurrent faculty attendance submissions at hourly bell transitions.	Supports 200 concurrent write operations without database contention.
NFR-AVAL-01	Availability	System uptime during institutional operating hours (07:00–20:00 IST).	99.9% Uptime (< 23 min unscheduled monthly downtime).
NFR-SEC-01	Security	Data in transit and data at rest encryption.	HTTPS (TLS 1.3 enforced) and AES-256 cloud encryption.

NFR ID	Category	Requirement Specification	Target Acceptance Metric
NFR-SEC-02	Security	Credential hashing and storage.	Argon2id or bcrypt (cost factor ≥ 12); no plaintext secrets.
NFR-RESP-01	Device Compatibility	Responsive layout adaptability across screen viewports.	Zero horizontal scrolling on viewports from 360px (mobile) to 2560px (4K).
NFR-BROW-01	Browser Support	Cross-browser client execution.	Chrome 90+, Edge 90+, Firefox 88+, Safari 14+ on Android, iOS, Windows, macOS.
NFR-ACC-01	Accessibility	Visual contrast and accessibility standards.	WCAG 2.1 Level AA compliance (minimum 4.5:1 text contrast).
NFR-DATA-01	Data Protection	Compliance with India DPDP Act and educational confidentiality.	Tenant data isolation; student personal records cannot leak across institutions.

13. INFORMATION ARCHITECTURE & KEY SCREENS

13.1 Site Structure & Navigation Hierarchy

- **Public Surface:** `/` (Landing Page) → `/login` (Unified Portal Login) → `/reset-password`
- **Student Surface:**
  - `/student/dashboard` (Overall percentage, quick stats, subject summary cards)
  - `/student/subjects/:id` (Subject detailed session history, presence records)
  - `/student/calculator` (Attendance recovery simulator)
  - `/student/profile` (Account information, password change)
- **Faculty Surface:**
  - `/faculty/dashboard` (Assigned classes, recent lectures, pending submissions)
  - `/faculty/mark` (Active roll-call interactive sheet)
  - `/faculty/history` (Past submitted sessions, 24h edit window)
  - `/faculty/reports` (Class cumulative percentage matrix, defaulters view)
- **Admin Surface:**
  - `/admin/dashboard` (Institutional macro-stats, missing lecture alerts)
  - `/admin/users` (Students and Faculty directory, CSV bulk import)
  - `/admin/academics` (Departments, Semesters, Sections, Subject mappings)
  - `/admin/reports` (Defaulters list generator, PDF/Excel export)
  - `/admin/audit-logs` (Immutable history of all system edits and overrides)

13.2 Key Screen Specifications

Screen Name	User Role	Core Functional Purpose	State Variants Handled
Digital Roll-Call Console	Faculty	Primary attendance input interface with one-tap status toggling and live count bar.	Loading, Populated Roster, Offline Cached, Submitting, Duplicate Warning, Success.
Student Attendance Hub	Student	Visual dashboard presenting subject percentage badges, shortage warnings, and deficit calculator.	Healthy (≥ 75%), At-Risk (70–74.9%), Debarred (< 70%), Empty (No lectures held).
Institutional Defaulter Engine	Admin	Filtering matrix to identify non-compliant students across departments for formal sanctions.	Default 75% Filter, Custom % Filter, Zero Defaulters Found, Exporting File.
Roster CSV Ingestion Tool	Admin	Upload interface for institution-wide bulk student/faculty provisioning.	Drag-and-Drop, Parsing, Error Preview Grid, Importing Progress, Completed.

14. HIGH-LEVEL DATA MODEL & ENTITY RELATIONSHIPS

The system utilizes a structured relational/document data model ensuring referential integrity across academic departments, classes, subjects, and attendance events.

14.1 Conceptual Entity Architecture

> **CORE ENTITY RELATIONAL MAPPING**
> **Institution** (1) → (N) **Department** (1) → (N) **ClassSection**
> **ClassSection** (1) → (N) **StudentEnrollment** (N) ← (1) **User (Student)**
> **Subject** (1) → (N) **AttendanceSession** (1) ← (1) **User (Faculty)**
> **AttendanceSession** (1) → (N) **AttendanceRecord** (N) ← (1) **User (Student)**
> **AttendanceSession** (1) → (N) **AuditLog** (Captures modifications)

14.2 Entity Catalog & Lifecycle Rules

Entity	Key Attributes	Relational Ownership	Lifecycle & Mutability Rules
User	`UserID, Email, PasswordHash, Role, FullName, Status`	Owned by Institution.	Created by Admin bulk import. Soft-deleted on graduation/departure.
ClassSection	`SectionID, DepartmentID, YearLevel, Semester, SectionName`	Belongs to Department.	Configured annually. Immutable once active attendance records are linked.
Subject	`SubjectID, SubjectCode, SubjectName, DepartmentID, Semester`	Belongs to Department.	Defined per curriculum. Cannot be dropped mid-term if records exist.

Entity	Key Attributes	Relational Ownership	Lifecycle & Mutability Rules
AttendanceSession	`SessionID, SubjectID, SectionID, FacultyID, Date, TimeSlot, Timestamp`	Created by Faculty.	Created on roll-call. Modifiable by Faculty for 24h; locked thereafter.
AttendanceRecord	`RecordID, SessionID, StudentID, Status (Present/Absent), ModifiedFlag`	Child of AttendanceSession.	Written atomically during session creation. Updates append audit logs.
AuditLog	`LogID, EntityType, EntityID, Action, ChangedBy, OldValue, NewValue, Timestamp, Reason`	System-wide governance.	Strictly append-only. Zero deletion or update permissions permitted for any role.

15. ROLES & PERMISSIONS (RBAC MATRIX)

The system enforces strict multi-tenant Role-Based Access Control. Permissions are evaluated on every server request.

Resource / Capability	Student Role	Faculty Role	Administrator Role
Own Attendance Overview	Read Only	Read Only	Read Only
Class Attendance Ledgers	No Access (403)	Read (Assigned Classes)	Read (All Department Classes)
Create Attendance Session	No Access (403)	Create (Assigned Subjects)	Create (Any Subject)
Edit Session (< 24h)	No Access (403)	Update (Own Sessions with Reason)	Update (Full Override)
Edit Session (> 24h)	No Access (403)	No Access (Locked)	Update (Mandatory Justification)
Bulk Import Users (CSV)	No Access (403)	No Access (403)	Full Create / Manage
Export Compliance Reports	No Access (403)	Read / Export (Own Subjects)	Full Read / Export (Institution-wide)
View Audit Log Trail	No Access (403)	No Access (403)	Read Only (Full Audit View)

16. INTEGRATIONS & THIRD-PARTY SERVICES

Service / Technology	Operational Purpose	Criticality	Degradation & Failure Fallback
Firebase / Firestore	Primary database engine for student identity, sessions, and records.	Mission-Critical	If unavailable, web application returns HTTP 503 with polite *"System undergoing database maintenance"* message.
Cloud Hosting (Render/AWS)	Hosts containerized Python Flask WSGI application server.	Mission-Critical	Auto-restarts failed worker processes. Health check endpoint at `/healthz` continuously probed.
SMTP Email Service	Dispatches password reset tokens and daily missing attendance reminders.	Non-Critical	Failure queued in background spooler; does not block core attendance marking workflows.
Client Device Storage	Browser local storage for session caching during intermittent Wi-Fi drops.	High	If disabled by private browsing, roll-call defaults to standard synchronous client-server POST.

17. ANALYTICS & INSTRUMENTATION

To evaluate user engagement, system health, and adoption bottlenecks, the application emits standardized telemetry events.

Event Identifier	Trigger Point	Payload Properties	Analytical Intent
`user_login_success`	Successful credential authentication.	`user_id, role, device_type`	Daily Active User (DAU) tracking.
`attendance_sheet_opened`	Faculty loads roll-call roster.	`faculty_id, subject_id, student_count`	Measures lecture initiation frequency.
`attendance_submitted`	Faculty clicks final submit confirmation.	`session_id, present_count, absent_count, duration_seconds`	Measures roll-call completion speed.
`attendance_session_edited`	Historical attendance modified.	`session_id, modified_by, hours_elapsed`	Monitors error rates and corrections.
`student_report_viewed`	Student views subject ledger.	`student_id, current_percentage`	Evaluates student self-service engagement.
`defaulter_report_exported`	Admin exports Excel/PDF list.	`admin_id, department, format`	Audits regulatory compliance generation.

18. AUDIT & OPERATIONAL LOGGING

Educational compliance demands complete non-repudiation. Every modification to an attendance ledger after initial submission is irreversibly cataloged.

> **AUDIT LOG IMMUTABILITY POLICY**
> Audit records are strictly insert-only. No user role—including the Institutional Super Administrator—possesses the capability to delete, truncate, or alter audit log rows. Each row records: `Timestamp (UTC), Actor ID, IP Address, Action Type, Target Record ID, Previous State, New State, Justification Text.`

Logged Action	Responsible Actor	Captured Data State	Retention & Inspection Rule
Attendance Modification	Faculty or Admin	Previous Status (e.g., Absent), New Status (Present), Justification Note.	Retained for minimum 5 academic years for statutory audit.
Account Role Change	Super Admin	Target User ID, Previous Role, Elevated Role.	Permanent retention; immediate admin security alert.
Bulk Data Deletion	Super Admin	Scope of records removed (e.g., archived graduating class).	Permanent retention; requires dual-admin sign-off.

19. RELEASE PLAN & PHASED ROADMAP

Phase	Theme	Core Scope Deliverables	Milestone Exit Criteria
Phase 0	Foundations & Architecture	Flask core boilerplate, Firebase schema setup, authentication, RBAC middleware, Docker containerization.	100% passing unit tests on auth; automated deployment pipeline live.
Phase 1 (MVP)	Core Operational Loop	Responsive roll-call UI, student self-service dashboards, batch CSV user ingestion, defaulters PDF export.	Pilot deployment across 3 engineering colleges; > 1,000 lectures logged.
Phase 2	Resilience & Engagement	Offline roll-call caching (Service Worker), automated email notifications for defaulters, student recovery calculator.	Zero attendance data lost during network drops; 90% faculty WAU.
Phase 3	Enterprise Scale	Multi-college SaaS tenant management, single sign-on (Google/Microsoft Workspace), dynamic timetable API integrations.	SaaS subscription billing engine active; multi-campus onboarding.

20. MINIMUM VIABLE PRODUCT (MVP) DEFINITION

> **MVP LAUNCH GATE CRITERION**
> *"The Attendora MVP is complete when a faculty member can open the application on any mobile smartphone browser, mark and submit attendance for a 60-student class in under 60 seconds, have those records instantly compute student percentages, and enable college administrators to export a verified low-attendance defaulters report."*

20.1 Feature Prioritization (MoSCoW Alignment)

- **Must Have (P0 - MVP Gate):** Secure RBAC login; interactive mobile roll-call with "Default Present"; real-time percentage calculator; student subject-wise view; CSV roster import; low-attendance filter (< 75%); PDF report export; 24h faculty modification grace period.
- **Should Have (P1 - Fast Follow):** Missing lecture alerts; deficit recovery calculator; date-range custom filtering; student dispute request button; institutional analytical overview dashboard.
- **Could Have (P2 - Post-Pilot):** Offline PWA capabilities; WhatsApp/SMS low-attendance notifications to parents; automated calendar timetable sync.
- **Won't Have (Excluded):** Biometric fingerprint hardware support; campus gate turnstile hardware integration; exam score and gradebook tracking.

21. RISKS & MITIGATIONS

Risk Category	Identified Risk Event	Severity	Engineered Mitigation Strategy
User Adoption	Faculty resist digital shift; find opening browser slower than paper.	High	Engineered "Default All Present" mechanism reduces roll-call to 4–5 screen taps total. In-person 15-minute training session for departmental faculty.
Infrastructure	Campus Wi-Fi connectivity blackouts inside basement classrooms.	High	Application designed with lightweight client-side payload (< 150KB); client local storage caches roll-call until connectivity is regained.
Data Integrity	Faculty marks wrong lecture slot or records proxy for absent student.	Medium	Duplicate-slot warning prompts; 24-hour modification lock; complete audit trail captures any status modification with mandatory written reason.
Security / Abuse	Students attempt credential brute-forcing to alter records.	Critical	Role validation executed strictly on server backend; client input sanitized against SQLi/XSS; rate limits on login routes (5 attempts / 10 min).
Cloud Cost	Database read/write spikes exceed free tier thresholds rapidly.	Medium	Denormalized cumulative attendance aggregates stored on student records; eliminates need to query hundreds of individual raw session records per page load.

22. OPEN QUESTIONS & DECISION LOG

#	Open Architectural / Business Question	Assigned Owner	Resolution Date	Current Working Assumption
OQ-01	Should low-attendance warnings be dispatched via SMS/WhatsApp directly to parents?	Product Lead / Dean Advisory	Pre-Phase 2	Excluded from MVP due to third-party SMS gateway costs; dashboard alerts used initially.

#	Open Architectural / Business Question	Assigned Owner	Resolution Date	Current Working Assumption
OQ-02	What is the exact institutional retention policy for graduating student records?	Compliance / Legal Lead	Phase 2 Launch	Records archived to cold storage after 4 academic years; audit logs retained indefinitely.
OQ-03	Should colleges be allowed to customize the defaulter threshold from 75% to other values (e.g., 80% for labs)?	Engineering Lead	Sprint 3	Yes; threshold stored as a configurable institutional settings variable (defaulting to 75.0%).

23. END-TO-END REQUIREMENT TRACEABILITY CHECK

To verify absolute consistency across this specification, all core business goals map directly through to technical execution:

- **Business Goal BG-02 (Faculty Retention)** → Product Goal 1 (Under 45s roll-call) → User Story (Bulk Mark Present) → **FR-FAC-02** → Verified by Acceptance Criteria → Prioritized as `P0` → Release Phase 1 (MVP).
- **Business Goal BG-01 (Institutional Onboarding)** → Product Goal 3 (Instant Auditing) → User Story (Defaulters Export) → **FR-ADM-03** → Verified by Acceptance Criteria → Prioritized as `P0` → Release Phase 1 (MVP).
- **Student Transparency Goal** → Product Goal 2 (Real-time visibility) → User Story (Shortage Calculator) → **FR-STU-03** → Verified by Acceptance Criteria → Prioritized as `P1` → Release Phase 2.

24. GLOSSARY OF TERMS & ACRONYMS

Term / Acronym	Operational Definition in Attendora Context
Roll-Call	The formal academic procedure of verifying and registering the physical presence or absence of enrolled students during an instructional session.
Defaulter	A student whose cumulative attendance percentage falls below the mandatory statutory institutional threshold (standard benchmark: 75.0%).
RBAC	Role-Based Access Control: Security mechanism restricting system access permissions strictly to authorized roles (Student, Faculty, Administrator).
VRSL	Verified Recorded Student-Lectures: The core North Star metric measuring active system throughput and institutional value delivery.
Idempotency	An API characteristic ensuring that identical duplicate attendance submissions (e.g., from network lag or double-clicking) cause zero data duplication.
BYOD	Bring Your Own Device: Operating model where teachers use their own smartphones or laptops to access the web application without dedicated college hardware.